

Code-Layout, Readability and Reusability

Date	7 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – Crop Recommendation System
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation and formatting followed throughout the Python and HTML code.
2	Proper File Structure	Files and folders are logically organized	Yes	Project organized into folders such as templates, static, datasets, and application files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables like temperature, humidity, prediction, and rainfall clearly describe their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as home() and prediction-related methods have descriptive names.
5	Code Comments	Inline and block comments explain logic	Yes	Comments are included to explain configuration, model loading, routes, and prediction logic.
6	Modular Design	Code is split into reusable functions/modules	Yes	Flask routes, model loading, and prediction logic are separated into reusable modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code is minimized, and reusable dictionaries/functions are

				used.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Exception handling (try-except)

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Flask Application (app.py)	Python, Flask	Handles routing and request processing	High
2	Crop Prediction Model	Scikit-learn, Joblib	Used to predict suitable crops	High
3	Input Validation Module	Python	Validates user input before	High
4	Crop Information Dictionary	Python	Displays crop descriptions after prediction	Medium
5	HTML Templates (index.html)	HTML, CSS, Bootstrap	User Interface for data input and displaying results	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project with proper folder structure.
Readability	5	Easy-to-read code with meaningful names and comments.
Reusability	5	Components are modular and can be reused in future enhancements.
Documentation / Comments	4	Code is documented with sufficient comments for understanding.
Overall Score	5/5	Clean, maintainable, and reusable machine learning web application.